Community detection in geometric graphs: Recent advances until 2026

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Résumé TODO : introduction UGCM, presenter la litterature

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1 Introduction

Community detection consists in finding clusters in a graph where nodes share more connections with nodes of the same cluster than with nodes of other clusters. This broad topic is detailed in [Fortunato, 2010]. In this report, we focus on random geometric graphs (RGG). Adding spatial information allows us to consider local information based on real or abstract proximity. In social networks, two individuals are more likely to be friends if they are geographically close, or if they share similar interests, age, etc. More concrete examples of RGG can be found in wireless networks [Gomez et al., 2015], in biology [Higham et al., 2008], and also in community detection problems [Sankararaman et al., 2020] word embedding, physic, ...

The main reference model for community detection is the Stochastic Block Model (SBM) [Holland et al., 1983]. Each node carries a hidden community label and edge probabilities depend on whether two nodes belong to the same community or not. This model comes from the Erdős-Rényi graph, where each node connects to others with the same probability [Erdős et al., 1960]. This model is now well understood, see [Abbe, 2018]. Nonetheless, it has some drawbacks, since it is unrealistic to think that every node can connect to every other one with the same probability scale. Around the same time as Erdős-Rényi, the RGG was introduced by [Gilbert, 1961]. Adding geometric information allows us to mimic some emergent behaviours of real networks, such as sparsity [Del Genio et al., 2011], the "small-world" property [Watts and Strogatz, 1998] or transitivity, namely the idea that a friend of a friend is more likely to be a friend.

RGG and its properties are well studied, see [Penrose, 2003] and more recently [Duchemin and De Castro, 2023]. When we add community structure, we are in a different setting. We observe the graph, and sometimes also the node locations, while the community labels remain hidden. The natural question is then whether the latent structure can be detected, and if so, how accurately the labels can be recovered. We are interested in several levels of inference. First, *distinguishability* asks whether the observed graph comes from a geometric model with communities or from a null model without community structure. This is a testing problem. It aims to distinguish the presence of a community structure, not to recover the labels. Then, if recovery is possible, one can ask how accurate it can be.

- *Weak/partial recovery* : detecting communities better than random guessing
- *Almost exact recovery* : the fraction of misclassified vertices vanishes as the number of nodes grows.
- *Exact recovery* : all community labels are correctly recovered, up to permutation.

Thus, in this report, community detection is not only understood as producing a convenient partition of the observed graph, but rather as recovering latent community labels under a probabilistic geometric model.

Previously, we emphasized properties that RGG share with real networks due to the geometric information. However, one can also obtain transitivity, and surely more, with attribute-based models [Li et al., 2018], hypergraphs [Ruggeri et al., 2023] and even geometric hypergraphs [Kovács et al., 2025]. We will not consider these models here, in order to keep the focus on the geometric aspect. Moreover, among the problems mentioned above, one is *Distinguishability* which should not be confused with the *Detection Problem* that we do not study here. The latter usually asks for conditions on the parameters defining the RGG under which the model is distinguishable from an Erdős-Rényi graph. Hence, this is not directly a community question. See, [Bubeck et al., 2016] [Liu et al., 2022] [Mao et al., 2026]. Finally, at the beginning of the literature, some works were interested in recovering the latent position of nodes in the geometric space, and considered this as a community detection problem, [Valdivia and De Castro, 2019] [Eldan et al., 2022]. This is a different problem from the one we are interested in, and it's in some sense harder, because one wants to recover the position (a continuous label) of each node rather than a community (a discrete label). TODO : parler des gaussian-mixture model + trouver ref ; voir commentaire latex !

2 Problem setting and inference tasks

Formally, we observe a graph $G \coloneqq (V, E)$, where V is the vertex set and E is the set of edges between vertices. Throughout the report, n denotes the asymptotic scaling parameter. Depending on the model, the observed number of vertices may be either deterministic or random, but is of order n. We consider a simple undirected graph Une remarque là dessus ? Pourquoi non-orienté est classique comme hypothese ?, without self-loops. The vertices are partitioned into k communities $C_1, \ldots, C_k$, so $V = \bigsqcup_{i=1}^{k} C_i$. We denote by $\sigma^\star : V \to [k]$ the ground-truth community label function, defined by $\sigma^\star(v) = i$ whenever $v \in C_i$. The community structure is hidden, and our goal is to recover it from the observed data.

An estimator is therefore a function of the observations. Depending on the model, the observations may consist only of the graph G, or of the graph together with the node locations $X = (X_v)_{v \in V}$. Accordingly, we write the estimator as $\hat{\sigma} = \hat{\sigma}(G)$ or $\hat{\sigma} = \hat{\sigma}(G, X)$, and we view $\hat{\sigma} : V \to [k]$ as an estimated labeling. In general, the labels are only identifiable up to a relabeling of the communities. Indeed, permuting the names of the communities does not change the underlying partition of the vertex set. For this reason, the quality of an estimator is measured up to permutation. Denoting by $\mathfrak{S}_k$ the symmetric group on $[k]$, we define

$$A_n(\hat{\sigma}, \sigma^\star) \coloneqq \frac{1}{|V|} \max_{\pi \in \mathfrak{S}_k} \sum_{u \in V} \mathbf{1}\big\{\hat{\sigma}(u) = \pi(\sigma^\star(u))\big\}$$

Thus, $A_n(\hat{\sigma}, \sigma^\star)$ measures the fraction of correctly classified vertices after the best possible relabeling of the communities. We now give more formal definitions of the problems introduced in Section 1.

Distinguishability. Following [Sankararaman and Baccelli, 2018], distinguishability is a hypothesis testing problem. Let $(\mathbb{P}_n)_{n \geq 1}$ be the laws of a geometric random graph model with community structure and let $(\mathbb{Q}_n)_{n \geq 1}$ be the laws of a null geometric model without communities, both defined on the same observation space. Let Ω_n be the observation space. A test is a measurable function $\varphi_n : \Omega_n \to \{0, 1\}$ where $\varphi_n = 1$ means that we decide in favor of the community model. We say that $(\mathbb{P}_n)$ and $(\mathbb{Q}_n)$ are distinguishable if there exist tests $(\varphi_n)_{n \geq 1}$ and a constant $\gamma > 0$ such that, under the uniform prior on $\{\mathbb{P}_n, \mathbb{Q}_n\}$,

$$\mathbb{P}\big(\varphi_n \text{ makes the correct decision}\big) = \frac{1}{2}\,\mathbb{P}_n(\varphi_n = 1) + \frac{1}{2}\,\mathbb{Q}_n(\varphi_n = 0) \geq \frac{1}{2} + \gamma - o_n(1)$$

Weak/Partial recovery. We say that weak/partial recovery is achievable if the estimator performs better than random guessing, that's to say in the symmetric case, if there exists a sequence of estimators $(\hat{\sigma}_n)_{n \geq 1}$ and a constant $\alpha > 1/k$ such that,

$$\mathbb{P}\big(A_n(\hat{\sigma}_n, \sigma^\star) \geq \alpha\big) \xrightarrow[n \to \infty]{} 1$$

Almost exact recovery. We say that almost exact recovery is achievable if there exists a sequence of estimators $(\hat{\sigma}_n)_{n \geq 1}$ such that for every $\varepsilon > 0$,

$$\mathbb{P}\big(A_n(\hat{\sigma}_n, \sigma^\star) \geq 1 - \varepsilon\big) \xrightarrow[n \to \infty]{} 1$$

Exact recovery. We say that exact recovery is achievable if there exists a sequence of estimators $(\hat{\sigma}_n)_{n \geq 1}$ such that,

$$\mathbb{P}\big(A_n(\hat{\sigma}_n, \sigma^\star) = 1\big) \xrightarrow[n \to \infty]{} 1$$

Most works in the literature focus on the case $k = 2$. Although several ideas extend beyond this setting, this is also the framework that we adopt in the following, except when specified otherwise. In that case, it is convenient to encode the labels by $\{-1,+1\}$, instead of $[2]=\{1,2\}$. Then "up to permutation" reduces to a global sign flip.

Each vertex also belongs to a space $\mathcal{X}$. In the following, we specify each time whether this space is latent or observed. For theoretical analysis, $\mathcal{X}$ is often chosen for mathematical convenience. The Euclidean space $\mathbb{R}^{d-1}$ is a natural reference because of its symmetry, homogeneity, isotropy, and direct geometric interpretation. See [Abbe et al., 2021], Citer ceux qui sont euclidiens. In the following, d is an integer. We also encounter the torus, mainly to avoid boundary effects and to focus on the core ideas of the proofs, Todo citer. Lastly, for latent models, the high-dimensional unit sphere $\mathbb{S}^{d-1}$ is a standard reference space to model similarity, and it also provides convenient spectral properties [Castro et al., 2020]. A summary of theoretical works on $\mathbb{S}^{d-1}$ is given in [Duchemin and De Castro, 2023].

In Section 3 we present models used throughout the literature. Hence, we suggest a model to unify them while remaining close to the notation used in the papers. We call it the *Unified Geometric Community Model* (UGCM) and it is defined as,

$$\mathrm{UGCM}\left(\mathcal{X},\rho,W_n,\mu_n,Z,\pi,\mathcal{Y},K_n\right)$$

where $\mathcal{X}$ is the ambient space and ρ is its metric. $W_n\subset\mathcal{X}$ is the window where nodes are sampled at the scale n. μ_n governs the spatial sampling mechanism. Usually, the vertices are sampled either as n i.i.d. points in W_n with law μ_n or as a Poisson Point Process (PPP) on W_n with intensity measure μ_n. We denote by Z the label set, and by $\pi=(\pi_i)_{i\in Z}$ the distribution of the labels. For all $u\in V$ and independently, $\sigma^\star(u)\sim\pi$. $\mathcal{Y}$ the space of pairwise observation. Let $\mathcal{P}(\mathcal{Y})$ denote the set of probability distributions on $\mathcal{Y}$, we consider the kernel observation,

$$K_n:\mathcal{X}\times\mathcal{X}\times Z\times Z\to\mathcal{P}(\mathcal{Y})$$

and for each unordered pair of vertices u,v, we observe,

$$Y_{uv}\mid(X,\sigma^\star)\sim K_n\big(X_u,X_v,\sigma^\star(u),\sigma^\star(v)\big)$$

The kernel is the core of the model. We note that Soft-RGG introduced by [Penrose, 2016] and developed in [Duchemin and De Castro, 2023] can be seen as a special case of UGCM where there exists a function $f_n:\mathbb{R}^+\to[0,1]$ such that for every $x,y\in\mathcal{X}$ and $z,z'\in Z$, $K_n(x,y,z,z')=\mathrm{Ber}(f_n(\rho(x,y)))$. Moreover, a particular case of Soft-RGG are the usual random geometric graph, i.e. the hard-threshold special case, when f_n is an indicator function, i.e. there exists $r_n>0$ such that $K_n(x,y,z,z')=\delta_{\mathbf{1}\{\rho(x,y)\le r_n\}}$. Then, this framework also accommodates weighted graphs, since the observation space $\mathcal{Y}$ is not necessarily binary.

As we will see, several phase-transition phenomena depend on the connectivity scale of the graph, which is usually measured through the average degree. Let D_n denote the degree of a vertex chosen uniformly at random. We can distinguish three regimes of connectivity, depending on the growth of $\mathbb{E}(D_n)$ as

$n \to \infty$. The most realistic but also the most delicate regime is the *sparse regime*. Each vertex only carries a finite amount of local information, the expected degree remains bounded,

$$\mathbb{E}(D_n) = \Theta(1)$$

In the *logarithmic regime*, the graph becomes sufficiently connected for exact recovery to become meaningful,

$$\mathbb{E}(D_n) = \Theta(\log n)$$

Finally, the *dense regime* corresponds to the easiest setting, the graph carries a lot of information,

$$\mathbb{E}(D_n) = \Theta(n)$$

3 Geometric community models and recovery thresholds

Faire le rapport dans l'ordre chronologique je pense

— [Galhotra et al., 2018] (Geometric Block Model, GBM)
— [Sankararaman and Baccelli, 2018] & [Abbe et al., 2021] (Geometric SBM)
— [Chien et al., 2020]
— [Galhotra et al., 2020] (GBM)
— [Galhotra et al., 2023] (GBM)
— [Avrachenkov et al., 2024] (Geometric Kernel Block Model, GKBM)
— [Gaudio et al., 2024b] (Geometric SBM)
— [Gaudio et al., 2024a] (Geometric Hidden Community Model, GHCM)
— [Gaudio and Guan, 2025] (GHCM)
— [Gaudio and Jin, 2026] (GHCM)

3.1 Geometric Block Model (GBM)

Historically, the first random geometric graph for community detection is defined as the Geometric Block Model (GBM) in [Galhotra et al., 2018]. They focus on 2 symmetric communities. Its defined as a hard-RGG on the unit sphere on high dimension for the Euclidian distance where vertices are placed uniformly on it. For $d \in \mathbb{N}\backslash\{0,1\},\ r_{in}, r_{out} \in [-1,1]$ and $r_{ij} := r_{in}\delta_{ij} + r_{out}(1-\delta_{ij})$

$$\mathrm{GBM}(d,r) := \mathrm{UGCM}(\mathbb{S}^{d-1}, \|\cdot\|_2, \mathbb{S}^{d-1}, \mathrm{Unif}(\mathbb{S}^{d-1}), \{-1,+1\}, \mathrm{Ber}(1/2), \{0,1\}, K_n)$$

with $K_n(X_u, X_v, Z_i, Z_j) = \mathbf{1}\{\langle X_u, X_v\rangle \geq r_{Z_iZ_j}\}$. They specified $r_{in} > r_{out}$, if its equal we can't distinguish communities and we impose greater to get higher edge density inside communities than across the communities. The recovery is done only from the adjency matrix, so the space S^{d-1} is assumed latent. parler du régime (car omega log(n)/n), c'est quel régime pour moi, degré moyen ? parler de leur algorithme et résultat.

Second paragraphe : RAG, amélioration résultat [Galhotra et al., 2020] [Galhotra et al., 2023]

Troisième paragraphe : Active learning, approche originale et la seule actuelle. Pessimisme preuve théorique. [Chien et al., 2020]

3.2 Geometric Stochastic Block Model (GSBM)

[Sankararaman and Baccelli, 2018] [Abbe et al., 2021]

The *Geometric Stochastic Block Model* (GSBM), introduced in [Sankararaman and Baccelli, 2018] and further developed in [Abbe, 2018], extends the classical SBM by incorporating a latent geometric structure. The vertices are embedded in a Euclidean space $\mathbb{R}^d$, typically according to a Poisson point process, and each vertex is assigned a hidden label $\sigma^\star(u) \in \{-1, +1\}$. An edge between u and v is then drawn with a probability that depends both on their distance $r = \|X_u - X_v\|_2$ and on their labels : one typically writes $p_{\mathrm{in}}^{(n)}(r)$ if $\sigma^\star(u) = \sigma^\star(v)$, and $p_{\mathrm{out}}^{(n)}(r)$ otherwise, with in general $p_{\mathrm{in}}^{(n)}(r) > p_{\mathrm{out}}^{(n)}(r)$. So the model can be naturally written in the UGCM framework as

$$\mathrm{GSBM}(d, r) = \mathrm{UGCM}\big(\mathbb{R}^d, \|\cdot\|_2, W_n, \mu_n, \{-1, +1\}, \mathrm{Ber}(1/2), \{0, 1\}, K_n\big)$$

Here, $W_n \subset \mathbb{R}^d$ is the observation window and $\mu_n = \lambda \mathbf{1}_{W_n}(x)\, dx$ is the intensity measure of a Poisson point process on W_n. The observation kernel is Bernoulli and depends both on geometry and on the latent labels :

$$K_n(x, y, z, z') = \mathrm{Ber}\Big(p_{\mathrm{in}}^{(n)}(\|x - y\|)\mathbf{1}_{\{z=z'\}} + p_{\mathrm{out}}^{(n)}(\|x - y\|)\mathbf{1}_{\{z\neq z'\}}\Big)$$

The foundational works of Sankararaman [Sankararaman and Baccelli, 2018] and [Abbe et al., 2021] show that geometry profoundly alters the recovery thresholds. In the sparse regime, $\mathbb{E}[D_n] = \Theta(1)$, each vertex carries only a finite amount of local information, which makes partial recovery difficult. In the logarithmic regime, $\mathbb{E}[D_n] = \Theta(\log n)$, exact recovery becomes the natural objective. In this setting, they obtain the necessary condition

$$\lambda \nu_d \Big(1 - \sqrt{ab} - \sqrt{(1 - a)(1 - b)}\Big) < 1$$

where λ is the intensity of the Poisson point process, ν_d is the volume of the unit ball in $\mathbb{R}^d$, and a, b are the within- and between-community connection parameters in the simple threshold model. They also introduce the *Good-Bad-Grid* algorithm, which partitions the space into cells, identifies sufficiently regular regions, and then propagates community information locally.

More recently,[Gaudio et al., 2024b] showed that this threshold can actually be achieved by a linear-time algorithm built in two stages : a local propagation step producing an almost-exact labeling, followed by a vertex-wise refinement based on a Poisson test. Finally, [Gaudio et al., 2024b] extended this perspective to more general Euclidean random graphs, in which the GSBM appears as a special case, and expressed the exact recovery threshold in terms of a Chernoff–Hellinger divergence.

3.3 (Geometric Kernel Block Model, GKBM)

[Avrachenkov et al., 2024] [Allem et al., 2025]

3.4 Geometric Hidden Community Model (GHCM)

[Gaudio et al., 2024b] [Gaudio et al., 2024a] [Gaudio and Guan, 2025] [Gaudio and Jin, 2026]

4 Spectral

(A confirmer) [Chen et al., 2016]

(papier SBM mais théorie intéressante) [Bordenave et al., 2015]

Cadre spectral intéressant pour certains espaces [Méliot, 2019]

Méthode spectrale + Robustesse (la perturbation est géométrique) [Péché and Perchet, 2020]

Méthode spectrale sur Soft Geometric Block Model (SGBM) (Generalisation de Geomemetric SBM) [Avrachenkov et al., 2021]

5 Ouvertures

- Considérer des modèles hyperboliques, pas seulement labels & distance-dependant, car ceux-ci captent mieux les propriétés habituelles des réseaux
  - Clustering [Krioukov, 2016]
  - Degre inhomogène [Barabási and Albert, 1999]
  - "Communities with specific mixing patterns" (mesoscale) [Toivonen et al., 2006]

  Communautés apparaissent dans un espace hyperbolique latent [Bruno et al., 2019] Réponse 6ans plus tard : pas toujours ! + definition d'un modele pour "régler" ça [Guarino et al., 2025]

  [Li et al., 2018]

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